Xinzhi (Berry) Liu, PhD

Power Electronics Researcher | GaN and SiC Device Reliability and Converter-Level Characterisation

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Right to work in the UK: Global Talent visa, no sponsorship required | IEEE Member

PROFILE

Postdoctoral researcher in power electronics with a hardware-first approach to wide bandgap devices. I build the test platforms, low-inductance PCBs and measurement workflows needed to characterise commercial GaN HEMTs and bidirectional switches under converter-representative conditions, and I use SPICE and TCAD to explain what the measurements show. First-author publications in *IEEE Transactions on Power Electronics* and *Microelectronics Reliability*; 15+ publications in total, including a joint paper with GE Vernova on SiC ruggedness. Earlier industry experience in substation protection design at Manitoba Hydro, Canada. Seeking a hardware or reliability engineering role in industry, or a further postdoctoral position in power semiconductor devices and converters.

Core expertise: GaN and SiC reliability testing (dynamic $R_{DS(on)}$, reverse conduction, threshold-voltage shift, short circuit); double-pulse and multi-pulse test design; high-frequency PCB and gate-drive design; measurement automation; SPICE and TCAD modelling.

PROFESSIONAL AND RESEARCH EXPERIENCE

Postdoctoral Research Fellow **Sep 2025 to Present**

School of Engineering, University of Warwick, UK

- Evaluate commercial 40 to 650 V GaN devices (Schottky p-GaN HEMTs, GITs, single-gate and dual-gate bidirectional switches including the Infineon CoolGaN GIT) under converter-representative stress: third-quadrant reverse conduction, dynamic on-resistance, threshold-voltage shift and short-circuit robustness.
- Designed converter-level single-shot and repetitive pulse test platforms with isolated stress control and programmable stress-release timing, giving non-destructive characterisation under realistic operating conditions.
- Designed a multi-board Altium adapter (motherboard and daughterboard) with dual isolated, Kelvin-referenced gate drivers and low-parasitic power and gate loops, to interface SMD bidirectional switches for UIS, surge, and short-circuit test; specified differential on-state voltage measurement for floating devices.
- Automated instrument control and data acquisition (Keithley 2651A and 2602B SMUs, oscilloscopes) and batch post-processing to extract knee voltage, dynamic reverse-conduction resistance, conduction-loss penalty and V_{TH} shift.
- Built LTspice electro-thermal models and Sentaurus TCAD simulations to interpret degradation signatures and support experimental validation.
- Contribute to collaborative studies on 1,200 V SiC linear-mode robustness, IGBT, SiC and GaN hybrid switches, and short-circuit, UIS and surge-current ruggedness.
- Appointed on the UKRI-funded project *Circular Microgrids: Circular Economy Pathways for Renewable Microgrids in Africa* (UKRI-320). Supervisor: Prof. Layi Alatise.

Visiting PhD Researcher **Sep 2023 to Mar 2025**

Southern University of Science and Technology (SUSTech), Shenzhen, China, supervised by Prof. Mengyuan Hua

- Designed multi-layer, high-frequency GaN HEMT converter PCBs, optimising switching performance, parasitics and thermal behaviour.
- Developed a multi-pulse test platform to extract dynamic $R_{DS(on)}$ and C_{OSS} under high-speed switching; this work led to my first-author *IEEE Transactions on Power Electronics* paper.
- Combined SPICE-based thermal and loss simulation with experimental validation of junction temperature and switching loss; used Keysight B1506 analyser for on-board Q_G-V_{GS} device characterisation.

PhD Research Assistant **Sep 2022 to Aug 2025**

Universiti Putra Malaysia, ALPER Centre, Malaysia

- Investigated dynamic on-resistance, charge trapping, gate reliability and switching oscillations in Schottky p-GaN HEMTs using multi-pulse test platforms and SPICE-assisted analysis.
- Designed and evaluated bus-transient fault-protection circuits in LTspice for bootstrap-driven GaN half-bridges and LLC converters, including sensing, comparator and shutdown stages.
- Combined MATLAB/Simulink PV MPPT control with LTspice device and switching-loss studies to compare GaN, silicon and SiC converters.
- Contributed to EMT modelling and validation of UPM's 11 kV/400 V distribution network to assess EV fast charging, PV integration and harmonic distortion.
- Supported solar-cell collaborations and UPM–Kyutech laboratory workshops; developed practical skills in Cadence analog IC layout and familiarity with high-voltage and partial-discharge testing.

Technical Support (Substation Design) **Apr 2019 to Feb 2020**

Manitoba Hydro, Manitoba, Canada

- Designed and drafted the upgrade of the B69R remote protection system at Raven Lake Substation and contributed to the B70H transfer-trip receiver system at Virden West Substation.
- Prepared the decommissioning and recovery plan, with engineering drawings, for a 4 kV transformer at Portage Saskatchewan Ave Substation.
- Compiled commissioning procedures and certification records and loaded drawings into the Design and Construction Office (DCO) system for traceability and regulatory compliance.

EDUCATION

PhD in Electrical Power Engineering **Sep 2022 to Aug 2025**

Universiti Putra Malaysia (UPM), Selangor, Malaysia

- Thesis focus: dynamic $R_{DS\,(on)}$ characterisation and multi-pulse C_{OSS} extraction in Schottky p-GaN HEMT power converters. Ranked 1st in Faculty (GPA 4.0/4.0). Best PhD Dissertation Award, IEEE CASS Malaysia Chapter (2025).

MSc in Green Technology (Energy), Distinction **Aug 2021 to Nov 2022**

Hong Kong Baptist University (HKBU), Hong Kong SAR

- Distinction; Ranked 1st in program (GPA 3.8/4.0). Research on nanostructured electron transport layers and mechanical modelling for flexible perovskite solar cells.

Diploma in Electrical Engineering Technology **Apr 2017 to Feb 2020**

Red River College Polytechnic, Winnipeg, Canada

- Honours; GPA 4.31/4.5. Capstone: lithium-ion battery management system and charger (Arduino-based monitoring, custom charger PCB, 3D-printed enclosure).

SELECTED PUBLICATIONS

Full list: ORCID 0000-0002-8032-0965

Journal articles, first author

[1] Liu**, X.**, Chen, J., Wang, S., Wu, S., Jiang, Z., Radzi, M. A. M., Azis, N., Shafie, S., & Hua, M. (2026). Output capacitance extraction of p-GaN HEMTs under multi-pulse switching across variable frequencies and drain-bias stress. *IEEE Transactions on Power Electronics*, 41(6), 8927-8937. (IF 7.1, JCR Q1)

[2] Liu**, X.**, Shafie, S., Radzi, M. A. M., Azis, N., Abdul Karim, A. H., Norddin, N., & Lawal, I. (2025). Simulation-based evaluation of power efficiency and output capacitance in standalone PV MPPT buck converters using 200 V p-GaN HEMTs. *Journal of Electrical Engineering & Technology*, 20, 3875-3887. (IF 2.0, JCR Q3)

[3] Liu**, X.**, Hua, M., Shafie, S., Radzi, M. A. M., & Azis, N. (2024). SPICE modelling-assisted evaluation of dynamic on-resistance characterization in Schottky p-GaN HEMTs amid synchronous buck transient instabilities. *Computers & Electrical Engineering*, 118, 109410. (IF 5.5, JCR Q1)

[4] Liu**, X.**, Shafie, S., Radzi, M. A. M., Azis, N., & Abdul Karim, A. H. (2024). Modelling and mitigating oscillation in E-mode GaN HEMT: A simulation-based approach to parasitic inductance optimization. *Microelectronics Reliability*, 152, 115293. (IF 2.5, JCR Q3)

[5] Liu**, X.**, & Shafie, S. (2025). Nanostructure silver absorbance with polarized light spectroscopy (PLS) and mobile imaging. *Instrumentation Science & Technology*, 53(4). (IF 1.5, JCR Q4)

Journal articles, co-author

[6] Mahmood, A., Ayala, H., Wan, H., **Liu, X.**, et al. (2026). Linear mode robustness analysis of 1200 V silicon and SiC MOSFETs. *IEEE Transactions on Power Electronics*, early access. (IF 7.1, JCR Q1)

[7] Ayala, H., Gonzalez, J. O., Deb, A., Malik, A. H., Jahdi, S., Wan, H., **Liu, X.**, & Alatise, O. (2026). Correlations between body diode pulsed overcurrent safe operating area, short circuit, and avalanche ruggedness in SiC MOSFETs. *IEEE Transactions on Electron Devices*, 73(6), 3308-3320. (IF 3.6, JCR Q2)

[8] Karim, A. H. A., Kadir, M. Z. A. A., **Liu, X.**, Azis, N., Norsahperi, N. M. H., & Abda, Z. M. K. (2026). Electromagnetic transient simulation of EV fast charging on distribution networks: Comparative evaluation with PV integration. *Journal of Energy Storage*, 155, 121535. (IF 10.7, JCR Q1)

Conference papers, first author

[9] Liu**, X.**, et al. (2026). Evaluation of third quadrant dynamic on-resistance and thermal impedance of GaN e-HEMTs using converter level switching tests. *ECCE Europe 2026*, Valencia, Spain.

[10] Liu**, X.**, Shafie, S., Radzi, M. A. M., Azis, N., Norddin, N., Abdul Karim, A. H., & Lawal, I. (2025). SPICE simulation assisted thermal management for surface-mount GaN HEMTs: Experimental validation and performance insights. *IEEE PELS International Decentralized Energy Access Symposium (IDEAS)*, 1-6.

PRESENTATIONS AND PROFESSIONAL ACTIVITIES

- Poster, CPE Annual Conference 2026, Edinburgh: third-quadrant dynamic on-resistance and thermal impedance of GaN e-HEMTs from converter-level switching tests.
- Peer reviewer for *IEEE Transactions on Power Electronics, Computers & Electrical Engineering, Microelectronics Reliability*, *IEEE Open Journal of Power Electronics* and *IET Circuits, Devices & Systems*.
- Member, IEEE (PELS, CAS, EDS, IES and PES).

AWARDS AND HONOURS

- 2025 Best PhD Dissertation Award, IEEE Circuits and Systems Society (CASS) Malaysia Chapter
- 2025 IEEE Power Electronics Society (PELS) Student Travel Grant
- 2023 to 2024 SUSTech International Fellowship; UPM Graduate Student Scholarship (PhD)
- 2022 Scholastic Award, Hong Kong Baptist University
- 2018 National Award, Electric Mobility Canada Kia Student Competition (team award for EV concept design)

TECHNICAL SKILLS

Power electronics and test: double-pulse and multi-pulse testing; dynamic $R_{DS\,(on)}$, C_{OSS}, reverse-conduction and short-circuit characterisation; DC-DC converters (buck, LLC half-bridge); GaN and SiC gate-drive and protection circuits.

Instruments: high-bandwidth oscilloscopes and high-voltage probes, Keithley 2651A and 2602B SMUs, Keysight B1505B power device analyser, power analysers, waveform generators.

PCB and simulation: Altium Designer, OrCAD and Cadence; LTspice, SIMPLIS, MATLAB/Simulink; Ansys (Q3D, Maxwell, Fluent), COMSOL; Sentaurus TCAD.

Programming and automation: Python, C++, Linux shell scripting, instrument-control and data-processing scripts, PLC ladder logic.

Languages: English (near-native proficiency), Mandarin (native), Cantonese and Malay (basic).

Referees available on request.